

Apply filters to SQL queries

Project description

This project is from the Linux and SQL module of the Google Cybersecurity Certificate. The goal here is to investigate a few potential security issues that involve login attempts and employee machines. Within SQL we will use filters to retrieve records from the pretend organization in the employees and log_in_attempts tables. The sections below will cover the commands used to run SQL queries and retrieve information from the database.

Retrieve after hours failed login attempts

We will examine failed login attempts after 18:00.

```
SELECT *  
FROM log_in_attempts  
WHERE login_time > '18:00' AND success = 0;
```

Retrieve login attempts on specific date

To examine login attempts on the 8th or 9th.

```
SELECT *  
FROM log_in_attempts  
WHERE login_time = '2022-05-09' OR login_time = '2022-05-08';
```

Retrieve login attempts outside of Mexico

Using NOT to exclude Mexico from the search instead of listing out all of the individual places. Easier this way.

```
SELECT *  
FROM log_in_attempts  
WHERE NOT country LIKE 'MEX%';
```

Retrieve employees in Marketing

```
SELECT *  
FROM employees
```

```
WHERE department = 'Marketing' AND office LIKE 'East-%';
```

Retrieve employees in Finance or Sales

```
SELECT *  
FROM employees  
WHERE department = 'Sales' OR department = 'Finance';
```

Retrieve all employees not in IT

```
SELECT *  
FROM employees  
WHERE NOT department = 'Information Technology';
```

Summary

- Investigated login activity by finding failed attempts after 18:00, identifying logins on May 8–9, 2022, and excluding any from Mexico.
- Analyzed employee data by listing Marketing staff in East offices, locating Finance or Sales staff, and showing all employees outside the IT department.